

Block (2/3)

The Finite Element Method for Applications in Electrical Engineering

EE4375 - FEM For EE Applications

Domenico Lahaye

DIAM - Delft Institute of Applied Mathematics

Last updated September 21, 2026

Modeling of Permanent Magnet Machines

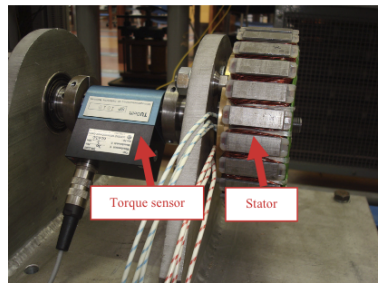


Fig. 2: Rotor of permanent magnet machine under study and test setup.

Modeling of Permanent Magnet Machines

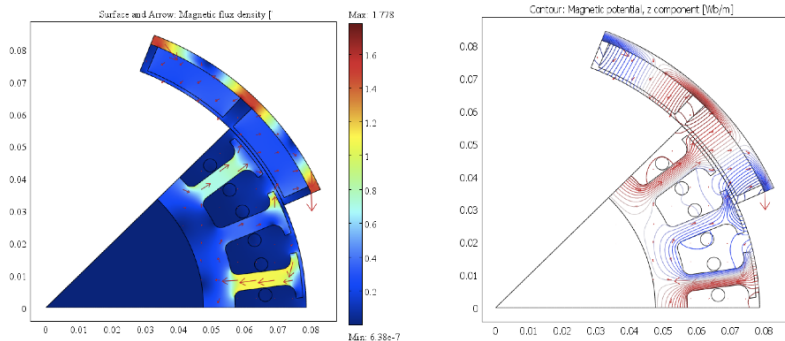


Fig. 6: Flux density and flux contour of the PM machine during load.

Modeling of Permanent Magnet Machines

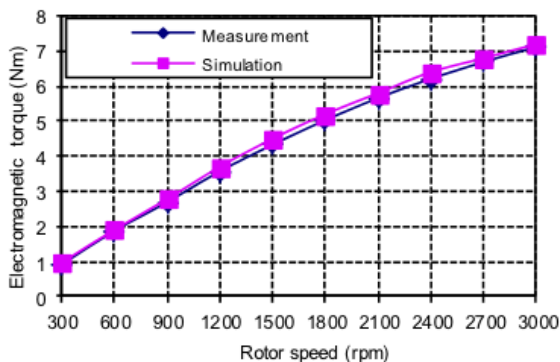
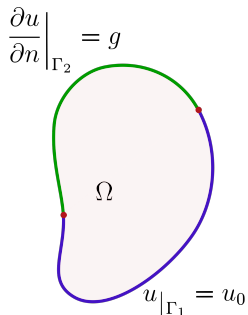


Fig.16. Mean electromagnetic torque vs. rotor speed.

Two-Dimensional Problem to Solve (1/4)

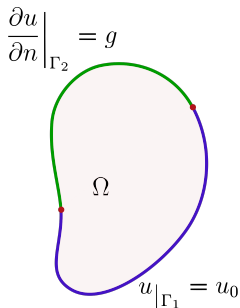
Geometry - Domain of Computation

- $(x, y) \in \Omega$ bounded domain in flat space



Two-Dimensional Problem to Solve (2/4)

Two Types of Boundary Conditions: $\Gamma = \Gamma_D \cup \Gamma_N$



- Dirichlet condition on Γ_D

fix u

(equivalent of $x = 0$ in 1D)

- Neumann condition on Γ_N

fix $\frac{\partial u}{\partial n}$

(equivalent of $x = 1$ in 1D)

Two-Dimensional Problem to Solve (3/4)

Boundary Value Problem for Second Order Differential Equations

- **given** $(x, y) \in \Omega$ with $\Gamma = \Gamma_D \cup \Gamma_N$ the boundary of Ω
- **given**: $f(x, y)$ given function and α given number
- **find**: $u(x, y)$ such that
 - $\Delta u(x, y) = f(x, y)$ for $(x, y) \in \Omega$ (differential equation on Ω)
 - $u = 0$ (Dirichlet boundary condition on Γ_D)
 - $\frac{\partial u}{\partial n} = \alpha$ (Neumann boundary condition on Γ_N)
- differential equation **invalid** on Γ - boundary conditions **valid** on Γ

Two-Dimensional Problem to Solve (4/4)

Residual function $r(x,y)$

- **definition:** $r(x,y) = \Delta u(x,y) + f(x,y)$
- **quality of the solution:**
small (large) residual is indication of good (poor) approximation
- **solve** – $\Delta u(x,y) = f(x,y) + \text{b.c.}$ equivalent to
- **find** $u(x,y)$ such that $r(x,y) = 0$ and $u(x,y)$ satisfies b.c.

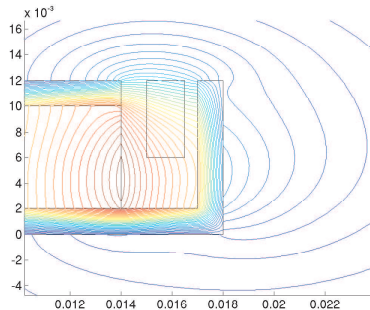
Mesh Generation (1/19)

Mesh Ω^h on Ω

- **nodes:** $\mathbf{x}_i = (x_i, y_i)$
- **edges:** edge_i
- **triangular elements:** e_i

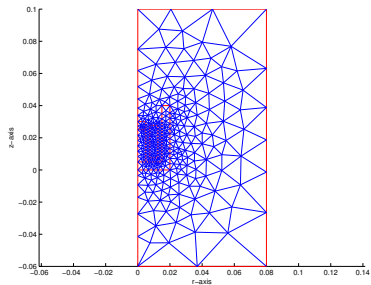
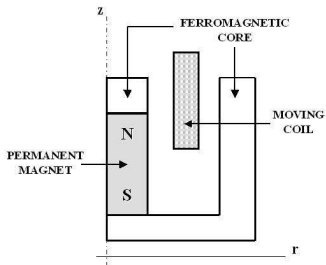
Mesh Generation (2/19)

Loudspeaker Example



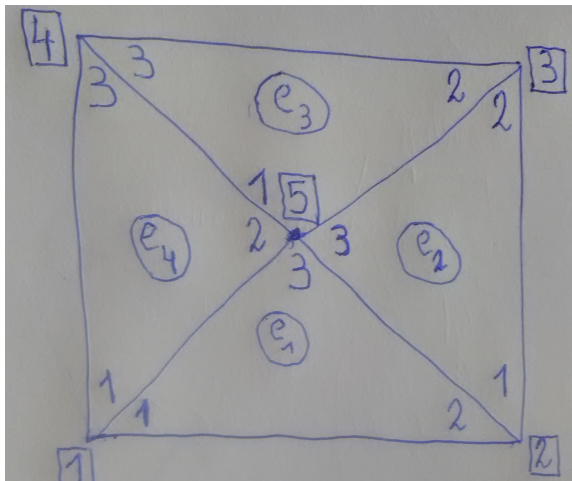
Mesh Generation (3/19)

Loudspeaker Example



Mesh Generation (4/19)

Four-Element Five-Node Mesh Example



Mesh Generation (5/19)

Four-Element Five-Node Mesh Example

Gmsh Output for First Order Elements

- **5 nodes**: 4 boundary nodes plus one interior node
- 8 edges (4 boundary edges plus 4 interior edges)
not reported separately by Gmsh
- **12 elements**: 4 triangles + 8 edges

Mesh Generation (6/19)

Local and Global Numbering of Nodes

- **local** numbering: numbering from 1 to 3 on each triangle e_i
- **global** numbering: numbering from 1 to n_{nodes} on the mesh Ω^h
- **local-to-global** mapping:
on each element e_i : given local number, find global number
- see Second Homework Assignment: matrix e
- valuable **bookkeeping tool** for assembly of matrix and vector

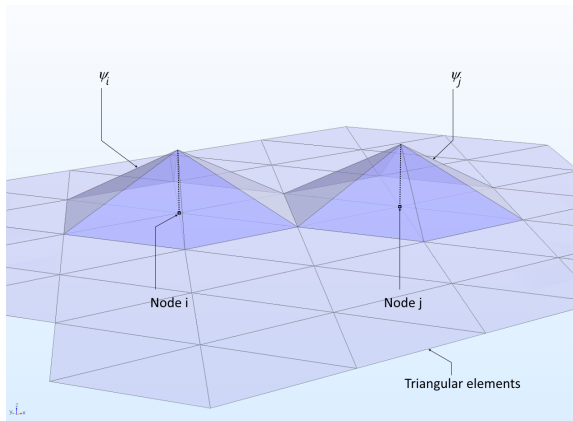
Mesh Generation (7/19)

Example of Local and Global Numbering of Nodes Four-Element Five-Node Mesh Example

- on element e_1
local nrs 1, 2 and 3 corresponds to global nrs 1, 2 and 5
- on element e_2
local nrs 1, 2 and 3 corresponds to global nrs 2, 3 and 5
- on element e_3
local nrs 1, 2 and 3 corresponds to global nrs 5, 3 and 4
- on element e_4
local nrs 1, 2 and 3 corresponds to global nrs 1, 5 and 4

Mesh Generation (8/19)

Shape Functions



Mesh Generation (9/19)

Definition of shape function $\phi_i(\mathbf{x}) = \phi_i(x, y)$

- linear Lagrange interpolation function $\phi_i(x, y)$
- linear means that $\phi_i(x, y) = C_1 x + C_2 y + C_3$
- each node $\mathbf{x}_i = (x_i, y_i)$ (including boundary nodes) has its $\phi_i(x, y)$
- $\phi_i(\mathbf{x}_j) = \delta_{ij}$ (see figure)

Mesh Generation (10/19)

What is the Buzz on Element e_i ?

- element e_i has three nodes $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ (local numbering)
- elements e_i "sees" three **linear** basis function (local numbering)

$$\phi_1(x, y) = a_1 x + b_1 y + c_1$$

$$\phi_2(x, y) = a_2 x + b_2 y + c_2$$

$$\phi_3(x, y) = a_3 x + b_3 y + c_3$$

- what are a_1 , b_1 and c_1 ? (same for $\phi_2(x, y)$ and $\phi_3(x, y)$)
- impose $\phi_1(\mathbf{x}_1) = 1$, $\phi_1(\mathbf{x}_2) = 0$ and $\phi_1(\mathbf{x}_3) = 0$
(same for $\phi_2(x, y)$ and $\phi_3(x, y)$)

Mesh Generation (11/19)

What is the Buzz on Element e_i ?

- three conditions $\phi_1(\mathbf{x}_1) = 1$, $\phi_1(\mathbf{x}_2) = 0$ and $\phi_1(\mathbf{x}_3) = 0$ read

$$\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

- same for $\phi_2(x, y)$ and $\phi_3(x, y)$

$$\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Important message: coordinate $x_1, \dots, y_3$ uniquely fix $a_1, \dots, c_3$!

Mesh Generation (12/19)

Exercise

- assume $\phi_i(x, y) = a_i x + b_i y + c_i$ for $1 \leq i \leq 3$
- compute $\nabla \phi_i(x, y)$ for $1 \leq i \leq 3$
- compute $\nabla \phi_i(x, y) \cdot \nabla \phi_j(x, y)$ for $1 \leq i, j \leq 3$
- compute $A_{e_k} = \int_{e_k} \nabla \phi_i(x, y) \cdot \nabla \phi_j(x, y) d\Omega$

Mesh Generation (13/19)

Exercise Solution

- assume $\phi_i(x, y) = a_i x + b_i y + c_i$ for $1 \leq i \leq 3$
- compute $\nabla \phi_i(x, y) = \left(\frac{\partial \phi_i(x, y)}{\partial x}, \frac{\partial \phi_i(x, y)}{\partial y} \right) = (a_i, b_i)$ for $1 \leq i \leq 3$
- compute $\nabla \phi_i(x, y) \cdot \nabla \phi_j(x, y) = \underbrace{a_i a_j + b_i b_j}_{\text{constant-in-}x\text{-and-}y}$ for $1 \leq i, j \leq 3$
- compute for $1 \leq i, j \leq 3$

$$A_{e_k} = \int_{e_k} \nabla \phi_i(x, y) \cdot \nabla \phi_j(x, y) d\Omega = \text{area}(e_k) [a_i a_j + b_i b_j]$$

- A_{e_k} : contribution to global matrix A on element e_k
- observe that A_{e_k} is independent of c_1, c_2 and c_3

Mesh Generation (14/19)

How to Compute the Elementary Matrix Contribution (1/2)

- define on each element the 3-by-3 matrix

$$Emat = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}$$

- compute $Emat$ by solving the linear system

$$\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} \underbrace{\begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{pmatrix}}_{=Emat} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- set third row of $Emat$ equal to zero or $Emat[3, :] = 0$

$$Emat = \begin{pmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ 0 & 0 & 0 \end{pmatrix}$$

Mesh Generation (15/19)

How to Compute the Elementary Matrix Contribution (2/2)

- then

$$Emat^T Emat = [a_i a_j + b_i b_j] \text{ for } 1 \leq i, j \leq 3$$

- and therefore contribution to global matrix A on element e_k

$$area(e_k) Emat^T Emat = area(e_k) [a_i a_j + b_i b_j] \text{ for } 1 \leq i, j \leq 3$$

Mesh Generation (16/19)

Computation of the Coefficients $a_1, \dots, b_3$ Revisited (1/4)

- motivation: solving the linear system for $Emat$ on each element likely computationally expensive
- alternative: solve symbolically using Cramer's Rule instead
- linear system for a_1 and b_1 (coefficient c_1 not required)

$$\underbrace{\begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}}_{=X} \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

•
$$a_1 = \frac{\det(X_1)}{\det(X)} \text{ and } b_1 = \frac{\det(X_2)}{\det(X)} \text{ and } c_1 \text{ not required}$$

Mesh Generation (17/19)

Computation of the Coefficients $a_1, \dots, b_3$ Revisited (2/4)



$$\begin{aligned}\det(X) &= \det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} = \det \begin{pmatrix} x_1 - x_3 & y_1 - y_3 & 0 \\ x_2 - x_3 & y_2 - y_3 & 0 \\ x_3 & y_3 & 1 \end{pmatrix} \\ &= (x_1 - x_3)(y_2 - y_3) - (x_2 - x_3)(y_1 - y_3) \\ &= 2 \text{ signed-area}(e_k)\end{aligned}$$

- $\bullet \det(X_1) = \det \begin{pmatrix} 1 & y_1 & 1 \\ 0 & y_2 & 1 \\ 0 & y_3 & 1 \end{pmatrix} = y_2 - y_3$

$$\det(X_2) = \det \begin{pmatrix} x_1 & 1 & 1 \\ x_2 & 0 & 1 \\ x_3 & 0 & 1 \end{pmatrix} = -(x_2 - x_3) = x_3 - x_2$$

Mesh Generation (18/19)

How to Compute the Coefficients $a_1, \dots, b_3$ (3/4)

- thus

$$a_1 = \frac{\det(X_1)}{\det(X)} = \frac{y_2 - y_3}{2 \text{ signed-area}(e_k)}$$

- and

$$b_1 = \frac{\det(X_2)}{\det(X)} = \frac{x_3 - x_2}{2 \text{ signed-area}(e_k)}$$

- similarly for a_2, a_3, b_2 and b_3

Mesh Generation (19/19)

How to Compute the Coefficients $a_1, \dots, b_3$ (4/4)

- local matrix A_{e_k} can be computed from nodal coordinates **without** solving 3-by-3 linear systems

Linear Combination of Shape Functions (1/3)

Linear Combination of Shape Functions

- set of functions $\{\phi_i(x) | 1 \leq i \leq n\}$ where $n = nnodes$
- **linear combinations** of these functions $\phi_i(x)$ can be made
- $V_0^h(\Omega)$: function space defined by all linear combinations

$$V_0^h(\Omega) = \text{span}\{\phi_1(x, y), \dots, \phi_n(x, y)\}$$

- $u^h(x, y) \in V_0^h(\Omega)$: there exists coordinates $c_1, \dots, c_n$ such that

$$u^h(x, y) = c_1 \phi_1(x, y) + \dots + c_n \phi_n(x, y)$$

Linear Combination of Shape Functions (2/3)

Application to Finite Elements

- $u(x, y)$: **exact** solution of the boundary value problem
- $u^h(x, y)$: finite element **approximation** to $u(x)$ computed on Ω^h
- $u^h(x, y) \in V_0^h(\Omega) = \text{span}\{\phi_1(x, y), \dots, \phi_n(x, y)\}$
- $u^h(x, y) = 0$ on Γ_D by definition of $V_0^h(\Omega)$
- expansion of $u^h(x)$ as linear combination of shape function

$$u^h(x, y) = c_1 \phi_1(x, y) + \dots + c_n \phi_n(x, y)$$

Linear Combination of Shape Functions (3/3)

Application to Finite Elements

- expansion of $u^h(x)$ as linear combination of shape function

$$u^h(x, y) = c_1 \phi_1(x, y) + \dots + c_n \phi_n(x, y)$$

- $c_1, \dots, c_n$ **coordinates** - $\phi_1(x, y), \dots, \phi_n(x, y)$ **basis functions**
- **basis functions**: unique determined by the mesh
- **coordinates**: to be determined by solving a linear system
one coordinate for each node $\mathbf{x}_i$ in the mesh

Strong vs. Weak Equal to Zero (1/2)

Weak or Variational Formulation

- n basis functions $\phi_i(x, y)$ defined by the mesh Ω^h
- inner product: $\langle g(x, y), \phi_i(x, y) \rangle = \int_{\Omega} g(x, y) \phi_i(x, y) d\Omega$
- $\langle g(x, y), \phi_i(x, y) \rangle$ coordinate of $g(x, y)$ along the basis function $\phi_i(x, y)$
- **numerically**: essential in remainder of the course

$$g(x, y) = 0 \text{ in discrete weak form}$$

$$\Leftrightarrow \forall \mathbf{x}_i \in \Omega^h : \langle g(x, y), \phi_i(x, y) \rangle = 0$$

n equations indexed by i where $n = n_{nodes}$

Strong vs. Weak Equal to Zero (2/2)

Applied to Finite Elements

- choose $g(x, y) = r(x, y) = \Delta u(x, y) + f(x, y)$
- enforce $g(x, y) = 0$ plus boundary conditions in discrete weak form

$$\langle g(x, y), \phi_i(x, y) \rangle = 0 \quad \text{for } 1 \leq i \leq n$$

$$\Leftrightarrow \int_{\Omega} g(x, y) \phi_i(x, y) d\Omega = 0 \quad \text{for } 1 \leq i \leq n$$

$$\Leftrightarrow \int_{\Omega} -\Delta u(x, y) \phi_i(x, y) d\Omega = \int_{\Omega} f(x, y) \phi_i(x, y) d\Omega \quad \text{for } 1 \leq i \leq n$$

where $n = n_{nodes}$

Calculus of functions in 2 vars (1/12)

Two Small Exercises

- recap: Block (2/3) of this course: $F(x) = v \frac{du}{dx}$
- suppose now that $\mathbf{F} = v \nabla u = v \operatorname{grad}(u)$
- Exercise 1: compute $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \nabla \cdot (v \nabla u)$
- Exercise 2: compute $\mathbf{F} \cdot \mathbf{n} = (v \nabla u) \cdot \mathbf{n}$

Calculus of functions in 2 vars (2/12)

Two Small Exercises: Exercise One

- $\nabla u = \text{grad}(u)$ is a **vector** function

$$\nabla u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right) = \frac{\partial u}{\partial x} \mathbf{i} + \frac{\partial u}{\partial y} \mathbf{j}$$

- $\mathbf{F} = v \nabla u$ is scalar times vector and thus again a **vector** function
(multiply each component of vector ∇u with scalar v)

$$\mathbf{F} = v \nabla u = v \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right) = \left(v \frac{\partial u}{\partial x}, v \frac{\partial u}{\partial y} \right)$$

- note that $\mathbf{F}$ is a **vector** function with two components

$$F_x = v \frac{\partial u}{\partial x} \quad \text{and} \quad F_y = v \frac{\partial u}{\partial y}$$

Calculus of functions in 2 vars (3/12)

Two Small Exercises: Exercise One

- $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \nabla \cdot (v \nabla u)$ is a **scalar** function

$$\begin{aligned}\operatorname{div} \mathbf{F} &= \nabla \cdot \mathbf{F} = \nabla \cdot \left(v \frac{\partial u}{\partial x}, v \frac{\partial u}{\partial y} \right) \\&= \frac{\partial}{\partial x} \left(v \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(v \frac{\partial u}{\partial y} \right) \\&= v \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} \right) \\&= v \Delta u + \nabla u \cdot \nabla v \\&= v \operatorname{Laplacian}(u) + \underbrace{\operatorname{grad}(u) \cdot \operatorname{grad}(v)}_{\text{inner-product-of-vectors}}\end{aligned}$$

Calculus of functions in 2 vars (4/12)

Two Small Exercises: Exercise 2

- $\mathbf{F} \cdot \mathbf{n}$ is the inner product of $\mathbf{F}$ and $\mathbf{n}$

$$\begin{aligned}\mathbf{F} \cdot \mathbf{n} &= (v \nabla u) \cdot \mathbf{n} \quad (\text{definition of } \mathbf{F}) \\ &= v \underbrace{(\nabla u \cdot \mathbf{n})}_{\text{inner-product-of-vectors}} \quad (\text{change order of operations}) \\ &= v \left(\frac{\partial u}{\partial x} n_x + \frac{\partial u}{\partial y} n_y \right) \quad (\text{inner product explicitly}) \\ &= v \frac{\partial u}{\partial n} \quad (\text{definition of the normal product})\end{aligned}$$

Calculus of functions in 2 vars (5/12)

Two Small Exercises: Summary

- suppose that $\mathbf{F} = v \nabla u$
- Ex 1: solution $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \nabla u \cdot \nabla v + v \Delta u$
- Ex 2: solution $\mathbf{F} \cdot \mathbf{n} = (v \nabla u) \cdot \mathbf{n} = v (\nabla u \cdot \mathbf{n}) = v \frac{\partial u}{\partial n}$
- these results will be used in the next two slides

Calculus of functions in 2 vars (6/12)

Recap: Integration of Function in One Variable

- assume $0 < x < 1$ or $x \in \Omega = (0, 1)$ and $F'(x) = \frac{dF(x)}{dx}$

$$\int_{\Omega} F'(x) dx = \underbrace{\int_0^1 F'(x) dx}_{1D\text{-line-integral}} = [F(x)]_0^1 = \underbrace{F(1) - F(0)}_{0D\text{-point-evaluation}}$$

- choose $F(x) = v(x) u'(x)$ (u has prime - v has no prime)
- arrive at integration by parts formula

$$-\int_0^1 u''(x) v(x) dx = \int_0^1 u'(x) v'(x) dx - [u'(x) v(x)]_0^1$$

- wish - goal - dream - ambition: repeat in 2D

Calculus of functions in 2 vars (7/12)

Integration by Parts in two variables

- Gauss Integration Theorem or Divergence Theorem

$(x, y) \in \Omega$ with boundary Γ

$$\underbrace{\int_{\Omega} \nabla \cdot \mathbf{F} d\Omega}_{2D\text{--surface--integral}} = \underbrace{\int_{\Gamma} \mathbf{F} \cdot \mathbf{n} ds}_{1D\text{--line--integral}}$$

- see calculus textbook or wiki
- choose: $\mathbf{F} = v \nabla u = (v \frac{\partial u}{\partial x}, v \frac{\partial u}{\partial y})$
- requires $\nabla \cdot \mathbf{F}$ and $\mathbf{F} \cdot \mathbf{n}$ from exercise

Calculus of functions in 2 vars (8/12)

Integration by Parts in two variables

- then

$$\begin{aligned}\int_{\Omega} \nabla \cdot \mathbf{F} d\Omega &= \int_{\Omega} \nabla \cdot (v \nabla u) d\Omega = \int_{\Omega} [\nabla u \cdot \nabla v + v \Delta u] d\Omega \\ &= \int_{\Gamma} \mathbf{F} \cdot \mathbf{n} ds = \int_{\Gamma} \frac{\partial u}{\partial n} v ds\end{aligned}$$

- after rearranging terms:

$$\int_{\Omega} (-\Delta u) v d\Omega = \int_{\Omega} \nabla u \cdot \nabla v d\Omega - \int_{\Gamma} \frac{\partial u}{\partial n} v ds$$

Calculus of functions in 2 vars (9/12)

Derivative: Integration by Parts in two variables

- integration by parts formula becomes

$$\int_{\Omega} (-\Delta u) v \, d\Omega = \int_{\Omega} \nabla u \cdot \nabla v \, d\Omega - \int_{\Gamma} \frac{\partial u}{\partial n} v \, ds$$

- observe that as before:

LHS: double derivatives in u - no derivatives on v

LHS: minus sign

RHS: first order derivatives on both u and v - additional term on the boundary

Calculus of functions in 2 vars (10/12)

Quadrature by Trapezoidal Rule

- trapezoidal rule: e_k triangle with vertices $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ (local numbering)

$$\int_{e_k} g(x, y) d\Omega \approx \frac{\text{area}(e_k)}{3} [g(\mathbf{x}_1) + g(\mathbf{x}_2) + g(\mathbf{x}_3)]$$

- $\text{area}(e_k)$ can be computed using $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ as input
- more accurate rules exist (e.g. Gauss quadrature, see references)

Calculus of functions in 2 vars (11/12)

Exercise

- assume that $g(x, y) = f(x, y) \phi_i(x, y)$ for $1 \leq i \leq 3$
- compute

$$\begin{pmatrix} \int_{e_k} f(x, y) \phi_1(x, y) d\Omega \\ \int_{e_k} f(x, y) \phi_2(x, y) d\Omega \\ \int_{e_k} f(x, y) \phi_3(x, y) d\Omega \end{pmatrix}$$

Calculus of functions in 2 vars (12/12)

Exercise Solution

- assume that $g(x, y) = f(x, y) \phi_i(x, y)$ for $1 \leq i \leq 3$
- compute

$$\begin{pmatrix} \int_{e_k} f(x, y) \phi_1(x, y) d\Omega \\ \int_{e_k} f(x, y) \phi_2(x, y) d\Omega \\ \int_{e_k} f(x, y) \phi_3(x, y) d\Omega \end{pmatrix} \approx \frac{\text{area}(e_k)}{3} \begin{pmatrix} f(\mathbf{x}_1) \\ f(\mathbf{x}_2) \\ f(\mathbf{x}_3) \end{pmatrix}$$

- contribution to global vector $\mathbf{f}$ on element e_k

Discrete Weak Form (1/6)

Apply Integration by Parts on Weak Formulation

- earlier we set the residual $r(x, y) = \Delta u''(x, y) + f(x, y)$ to zero in weak form and arrived at

$$\int_{\Omega} (-\Delta u) \phi_i(x, y) d\Omega = \int_{\Omega} f(x, y) \phi_i(x, y) d\Omega \quad \text{for all } 1 \leq i \leq n$$

- apply integration by part to the LHS

$$\int_{\Omega} \nabla u \cdot \nabla \phi_i(x, y) d\Omega = \int_{\Omega} f(x, y) \phi_i(x, y) d\Omega + \int_{\Gamma} \frac{\partial u}{\partial n}(x, y) \phi_i(x, y) ds$$

for all $1 \leq i \leq n$ where $n = \text{nnodes}$

observe: minus sign in LHS disappeared

- first order derivative on both $u(x)$ and $\phi_i(x)$

Discrete Weak Form (2/6)

- Dirichlet boundary conditions: $u = 0$ on Γ_D
- Neumann boundary conditions: $\frac{\partial u}{\partial n} = \alpha$ on Γ_N
- Dirichlet and Neumann boundary conditions treated differently
- boundary term in the RHS of the weak form

$$\begin{aligned} \int_{\Gamma} \frac{\partial u}{\partial n}(x, y) \phi_i(x, y) ds &= \int_{\Gamma_D} \frac{\partial u}{\partial n}(x, y) \phi_i(x, y) ds \\ &\quad + \underbrace{\int_{\Gamma_N} \frac{\partial u}{\partial n}(x, y) \phi_i(x, y) ds}_{=\alpha} \end{aligned}$$

- we thus obtain that

$$\int_{\Gamma} \frac{\partial u}{\partial n}(x, y) \phi_i(x, y) ds = \int_{\Gamma_D} \frac{\partial u}{\partial n}(x, y) \phi_i(x, y) ds + \int_{\Gamma_N} \alpha \phi_i(x, y) ds$$

Discrete Weak Form (3/6)

- Dirichlet boundary conditions: impose that $\phi_i(x, y) = 0$ on Γ_D
- we thus obtain that

$$\int_{\Gamma} \frac{\partial u}{\partial n}(x, y) \phi_i(x, y) ds = \int_{\Gamma_N} \alpha \phi_i(x, y) ds$$

- discrete weak form becomes

$$\int_{\Omega} \nabla u \cdot \nabla \phi_i(x, y) d\Omega = \int_{\Omega} f(x, y) \phi_i(x, y) d\Omega + \int_{\Gamma_N} \alpha \phi_i(x, y) ds$$

for all $1 \leq i \leq n$

Discrete Weak Form (4/6)

Discrete Weak Form Becomes for $1 \leq i \leq n$

$$\int_{\Omega} \nabla u(x, y) \cdot \nabla \phi_i(x, y) d\Omega = \int_{\Omega} f(x, y) \phi_i(x, y) d\Omega + \int_{\Gamma_N} \alpha \phi_i(x, y) ds$$

- assume $u(x, y)$ approximate by $u^h(x, y)$ where $u^h(x, y) = \sum_{j=1}^n c_j \phi_j(x, y)$
- thus $u'(x)$ approximate by $\nabla u^h(x, y) = \sum_{j=1}^n c_j \nabla \phi_j(x, y)$
- then for $1 \leq i \leq n$

$$\sum_{j=1}^n \int_{\Omega} \nabla \phi_j(x, y) \cdot \nabla \phi_i(x, y) dx c_j = \int_{\Omega} f \phi_i d\Omega + \int_{\Gamma_N} \alpha \phi_i(x, y) ds$$

Discrete Weak Form (6/6)

Discrete Weak Form Becomes

- for $1 \leq i \leq n$ where $n = nnodes$

$$\sum_{j=1}^n \int_{\Omega} \nabla \phi_j(x, y) \cdot \nabla \phi_i(x, y) d\Omega c_j = \int_{\Omega} f(x, y) \phi_i(x, y) d\Omega \\ + \int_{\Gamma_N} \alpha \phi_i(x, y) ds$$

- can be written in the form: for $1 \leq i \leq n$
index i counts equations - index j counts unknowns

$$\sum_{j=1}^n A_{ij} c_j = f_i$$

- and thus as a n by n linear system

$$A \mathbf{c} = \mathbf{f}$$

Linear System Formulation (1/3)

Expression for Matrix and Vector Elements

- Matrix elements:

$$A_{ij} = \int_{\Omega} \nabla \phi_j(x, y) \cdot \nabla \phi_i(x, y) d\Omega \text{ for } 1 \leq i, j \leq n$$

- Vector elements:

$$f_i = \int_{\Omega} f(x, y) \phi_i(x, y) d\Omega + \int_{\Gamma_N} \alpha \phi_i(x, y) ds \text{ for } 1 \leq i \leq n$$

Linear System Formulation (2/3)

Properties of Matrix A

- A is **large**
 $n > 1\text{e}6$ in 3D applications in no exception
- A is **sparse**
 A contains many zero elements (cfr. 2D finite difference method)
- A many other cool properties $\Rightarrow$ **fast solvers** for $A\mathbf{c} = \mathbf{f}$ exist

Linear System Formulation (3/3)

Summary: Matrix A and Right-Hand Vector $\mathbf{f}$

Treatment of the Boundary Conditions

- **Dirichlet boundary conditions:** $u(x, y) = 0$:

modify equations corresponding to the boundary nodes in linear system

see finite difference method

- **Neumann boundary conditions:** $\frac{\partial u}{\partial n} = \alpha$ on Γ_N

add term $\int_{\Gamma_N} \alpha \phi_i(x, y) ds$ to vector $\mathbf{f}$

see lab sessions for details

Element-by-Element Construction of the Vector (1/2)

How does element e_k contribute to the vector $\mathbf{f}$?

- $f_{e_k} \in \mathbb{R}^3$ contribution of element e_i to global vector $\mathbf{f}$
using local numbering of the three nodes on the element e_k

$$f_{e_k} = \begin{pmatrix} \int_{e_k} f(x, y) \phi_1(x, y) dx \\ \int_{e_k} f(x, y) \phi_2(x, y) dx \\ \int_{e_k} f(x, y) \phi_3(x, y) dx \end{pmatrix}$$

- use trapezoidal rule of integration (see earlier)

$$f_{e_k} = \begin{pmatrix} \int_{e_k} f(x, y) \phi_1(x, y) d\Omega \\ \int_{e_k} f(x, y) \phi_2(x, y) d\Omega \\ \int_{e_k} f(x, y) \phi_3(x, y) d\Omega \end{pmatrix} \approx \frac{\text{area}(e_k)}{3} \begin{pmatrix} f(\mathbf{x}_1) \\ f(\mathbf{x}_2) \\ f(\mathbf{x}_3) \end{pmatrix}$$

- given mesh Ω^h and source $f(x)$, f_{e_k} for each e_k can be computed

Element-by-Element Construction of the Vector $\mathbf{f}$ (2/2)

Finite Element Assembly of the Vector $\mathbf{f}$

- loop over all of the N elements e_k in the mesh Ω^h
- on e_k compute the local element vector $f_{e_k} \in \mathbb{R}^3$
the local element vector has **three** components
- add local element vector to the global vector $\mathbf{f} \in \mathbb{R}^n$
the global vector $\mathbf{f}$ has **n** components where $n = nnodes$
- $\mathbf{f} = \mathbf{f} + f_{e_k}$
assembly requires taking the mesh connectivity into account
connectivity here refers to mapping from local to global
numbering on the element e_k

Element-by-Element Construction of the Matrix (1/2)

How does element e_k contribute to the vector A ?

- $A_{e_k} \in \mathbb{R}^{3 \times 3}$ contribution of element e_j to global vector A

$$A_{e_k} = \left(\int_{e_k} \nabla \phi_i(x, y) \cdot \nabla \phi_j(x, y) d\Omega \right)_{1 \leq i, j \leq 3}$$

- using derivative of the shape functions (see earlier)

$$A_{e_k} = \text{area}(e_k) (a_i a_j + b_i b_j)_{1 \leq i, j \leq 3}$$

Element-by-Element Construction of the Matrix (2/2)

Finite Element Assembly of the Matrix A

- loop over all of the N elements e_k in the mesh Ω^h
- on e_i compute the local element matrix $A_{e_k} \in \mathbb{R}^{3 \times 3}$
the local element matrix has **three by three** components
- add local element matrix to the global matrix $A \in \mathbb{R}^{n \times n}$
the global matrix A has **n by n** components where $n = \text{nnodes}$
- $A = A + A_{e_k}$
assembly requires taking the mesh connectivity into account
connectivity here refers to mapping from local to global
numbering on the element e_k

Computation on Reference Element (1/8)

Area of Triangle in Mesh

- triangle $\mathbf{t}$ in mesh with nodes
 $\mathbf{x}_1 = (x_1, y_1)$, $\mathbf{x}_2 = (x_2, y_2)$ and $\mathbf{x}_3 = (x_3, y_3)$
- direction vectors $\mathbf{x}_{12} = \mathbf{x}_2 - \mathbf{x}_1$ and $\mathbf{x}_{13} = \mathbf{x}_3 - \mathbf{x}_1$
- area-triangle equal to $.5\|\mathbf{x}_{12} \times \mathbf{x}_{13}\|$

$$\bullet \mathbf{x}_{12} \times \mathbf{x}_{13} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_2 - x_1 & y_2 - y_1 & 0 \\ x_3 - x_1 & y_3 - y_1 & 0 \end{vmatrix} =$$
$$0\mathbf{i} + 0\mathbf{j} + [(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)]\mathbf{k}$$

- area-triangle
 $= .5\|\mathbf{x}_{12} \times \mathbf{x}_{13}\| = .5 |(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)|$

Computation via Ref Element (2/8)

Coordinate Transformation from Configuration Space (ξ, η)
to Physical Phase (x, y) (need figure)

- triangle $\mathbf{t}$ in mesh with nodes
 $\mathbf{x}_1 = (x_1, y_1)$, $\mathbf{x}_2 = (x_2, y_2)$ and $\mathbf{x}_3 = (x_3, y_3)$
- triangle $\hat{\mathbf{t}}$ in configuration space or (ξ, η) -space with nodes
 $(\xi, \eta) = (0, 0)$, $(\xi, \eta) = (1, 0)$ and $(\xi, \eta) = (0, 1)$
- mapping from reference space to physical space
$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x(\xi, \eta) \\ y(\xi, \eta) \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix}$$
- maps $(\xi, \eta) = (0, 0)$ to $(x, y) = (x_1, y_1)$,
 $(\xi, \eta) = (1, 0)$ to $(x, y) = (x_2, y_2)$ and
 $(\xi, \eta) = (0, 1)$ to $(x, y) = (x_3, y_3)$
- observe similarity with the one-dimensional case

Computation via Ref Element (3/8)

Inverse Coordinate Transformation and Jacobians

- mapping from reference space to physical space

$$\begin{pmatrix} \xi \\ \eta \end{pmatrix} = \begin{pmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{pmatrix}^{-1} \begin{pmatrix} x - x_1 \\ y - y_1 \end{pmatrix}$$

- Jacobian $J = \frac{\partial(x,y)}{\partial(\xi,\eta)} = \begin{pmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{pmatrix} = \begin{pmatrix} x_\xi & x_\eta \\ y_\xi & y_\eta \end{pmatrix}$

- determinant of the Jacobian

$$\det(J) = \left| \frac{\partial(x,y)}{\partial(\xi,\eta)} \right| = |(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)| \text{ or } \det(J) = 2 \text{ area-triangle}$$

- Jacobian of inverse transformation or inverse Jacobian

$$J^{-1} = \frac{1}{\det(J)} \begin{pmatrix} y_3 - y_1 & -(y_2 - y_1) \\ -(x_3 - x_1) & x_2 - x_1 \end{pmatrix}$$

Computation via Ref Element (4/8)

Basis Functions on Reference and Physical Element

- on the reference element: $\hat{\phi}_1(\xi, \eta) = \xi$
- on the physical element: $\phi_1(x, y) = \phi_1(x(\xi, \eta), y(\xi, \eta)) = \hat{\phi}_1(\xi, \eta)$
- similar for $\phi_2(x, y)$ and $\phi_3(x, y)$

Computation via Ref Element (5/8)

Chain Rule and Jacobian Transformations

- chain rule $\frac{\partial}{\partial x} = \frac{\partial}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial \eta}{\partial x} \frac{\partial}{\partial \eta}$
- thus $\nabla_{(x,y)} = J \nabla_{(\xi,\eta)}$

Computation via Ref Element (6/8)

Integration over mesh triangle $\mathbf{t}$ via Coordinate Transformation
and integration over $\hat{\mathbf{t}}$

- $\int_{\mathbf{t}} g(x, y) \, dx \, dy = \int_{\hat{\mathbf{t}}} g(x(\xi, \eta), y(\xi, \eta)) \left| \frac{\partial(x, y)}{\partial(\xi, \eta)} \right| \, d\xi \, d\eta$
- $\int_{\mathbf{t}} g(x, y) \, dx \, dy = 2 \text{ area-triangle} \int_{\hat{\mathbf{t}}} g(x(\xi, \eta), y(\xi, \eta)) \, d\xi \, d\eta$
- is it sufficient to have the Jacobian and the basis functions on the reference element ?

Computation via Ref Element (7/8)

Small Example - Proof of the Pudding



Computation via Ref Element (8/8)

Linear Basis Function on the Reference Element

- basis functions

$$\psi_1(\xi, \eta) = 1 - \xi - \eta, \psi_2(\xi, \eta) = \xi \text{ and } \psi_3(\xi, \eta) = \eta$$

- basis function gradients

$$\nabla \psi_1(\xi, \eta) = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \nabla \psi_2(\xi, \eta) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \nabla \psi_3(\xi, \eta) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

- basis function gradients pairwise inner products

$$\nabla \psi_i \cdot \nabla \psi_j = \begin{pmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \text{ for } 1 \leq i, j \leq 3$$

constant on the element

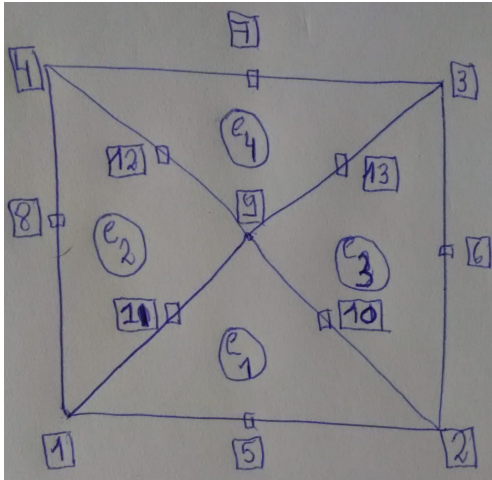
- integral over triangle of pairwise inner products - area reference

$$\text{triangle is } 0.5 - \int_{\text{triangle}} \nabla \psi_i \cdot \nabla \psi_j \, dx \, dy = 0.5 \begin{pmatrix} 2 & -1 & -1 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

$$1 \leq i, j \leq 3$$

Second Order Elements

Second Order Four-Element Five-Node Mesh Example



Exercises

- to be defined